Approximating a Nonlinear Function with Fuzzy Systems

Fuzzy Control, Project 1

Arshia Goshtasbi • MSc Electrical Engineering (Control), Iran University of Science and Technology • Autumn 1403 (2024)

Problem. Approximate

$$g(x) = 2\cos(2.5x) + \sin(4x) + 1.1\cos(0.3x), \qquad x \in [0, 25]$$

(sampled every 0.05) to accuracy $\varepsilon = 0.02$, first with designs that use the formula of g (required part) and then from data only (bonus part). Every fuzzy system here is written by hand (membership functions, product inference, centre-average defuzzifier); no toolbox is used.

Designs that use g. The input is partitioned by membership functions spaced h apart, and the rule outputs are set to g at the centres:

- *First-order accuracy:* $\|g - f\|_\infty \le \|g'\|_\infty h$, so $h = \varepsilon/\|g'\|_\infty = 0.02/8.89 = 0.00225$, i.e. 11,117 membership functions.
- *Second-order accuracy:* $\|g - f\|_\infty \le \frac{1}{8}\|g''\|_\infty h^2$, so $h = \sqrt{8\varepsilon/\|g''\|_\infty} = 0.0751$ with $\|g''\|_\infty = 28.37$, i.e. 333 membership functions.

Data-driven designs (from 200–500 samples of g):

- a lookup table built from the data, with one rule per sample and conflicting rules resolved by the highest degree (300 membership functions);
- a Gaussian fuzzy system with 50 rules whose centres, widths and outputs are trained by gradient descent (10,000 epochs);
- a nearest-neighbour clustering design with 200 clusters.

For comparison, recursive least squares (forgetting factor 0.999) was fitted on the basis $\{\cos 2.5x, \sin 4x, \cos 0.3x, 1\}$

Results (re-run in MATLAB R2023b):

Design	Rules / MFs	Max $\lvert g - f \rvert$	MSE
First order, triangular	11117	1.8×10^{-5}	4.4×10^{-11}
First order, Gaussian ($\sigma = h$)	11117	0.0056	5.4×10^{-8}
Second order, triangular	333	0.0199	5.5×10^{-5}
Second order, Gaussian	334	0.189	1.7×10^{-3}
Lookup table, triangular	300	0.104	9.5×10^{-4}
Lookup table, Gaussian	300	0.474	3.0×10^{-2}
Gradient descent, Gaussian	50	0.330	7.8×10^{-3}
Clustering	200	0.423	1.8×10^{-2}

The triangular designs meet ε, because the error bounds above are derived for triangular (piecewise-linear) partitions. The Gaussian first-order design also meets it, thanks to the very fine grid. The Gaussian second-order design does not: overlapping Gaussians don't interpolate the centre values exactly. The data-driven designs use far fewer rules but don't reach ε. Least squares recovers the coefficients $(2, 1, 1.1, 0)$ exactly, but only because it is given the true basis functions; it is a sanity check, not a fuzzy approximator.

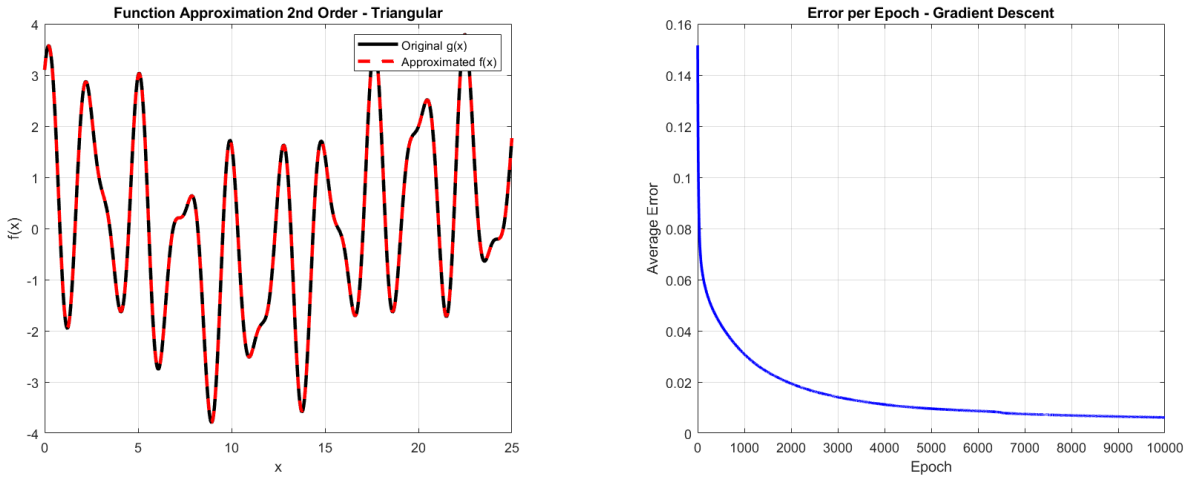


Figure 1: Second-order triangular design (left) and the training error of the gradient-descent design (right).